

# **Final Report: Context-Aware Spatiotemporal Traffic Risk Forecasting**

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## Abstract

This project builds a context-aware traffic risk forecasting system for Ottawa road segments. It began as a static collision map, but evolved into an interactive system that combines normalized spatial risk, weather-aware prediction, spatially valid evaluation, live inference, and local AI explanations. Users can load a neighborhood, score nearby roads under live or historical weather, and inspect why a segment is currently in a lower, moderate, or high risk band. The final model is evaluated with 2 km block spatial cross-validation and reported using ROC-AUC and Average Precision.

## 1. Introduction

Historical collision maps are useful, but they do not answer a more practical question: where are collisions more likely under current conditions? This project addresses that gap by building a road-segment risk forecasting system for Ottawa.

The final goal is not only to visualize past crashes, but to estimate current segment-level risk under a selected weather scenario, present the result on an interactive map, and explain the result in plain language.

This requires:

- geospatial data engineering
- road-segment feature construction
- weather-aware machine learning
- spatially valid evaluation
- real-time inference
- explainable frontend interaction

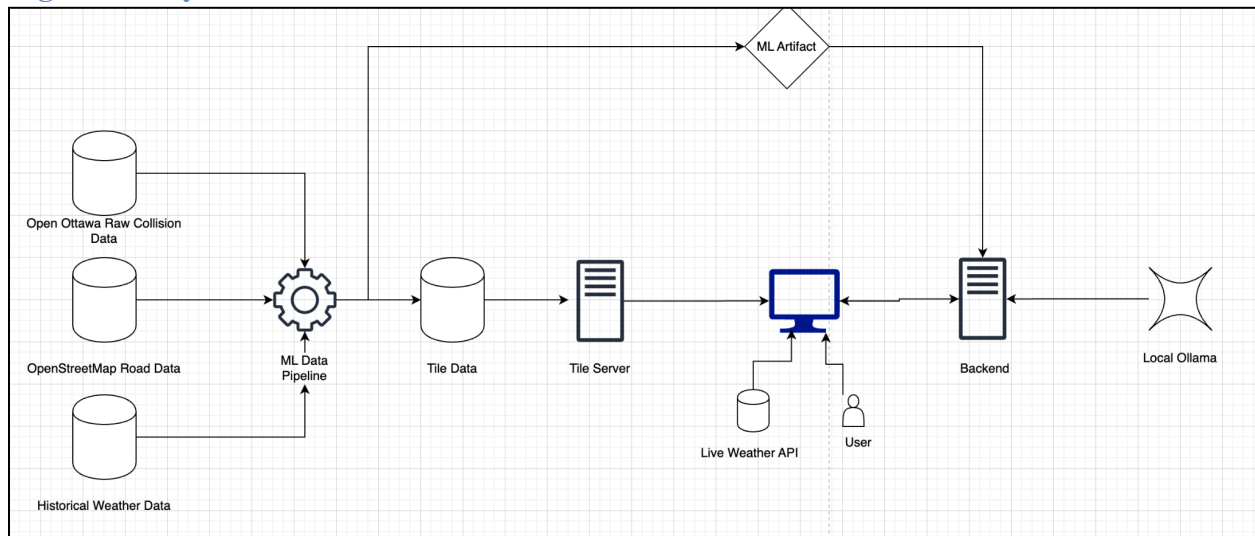
## 2. System Overview

The system predicts road-segment risk in Ottawa under either live weather or a user-selected historical date.

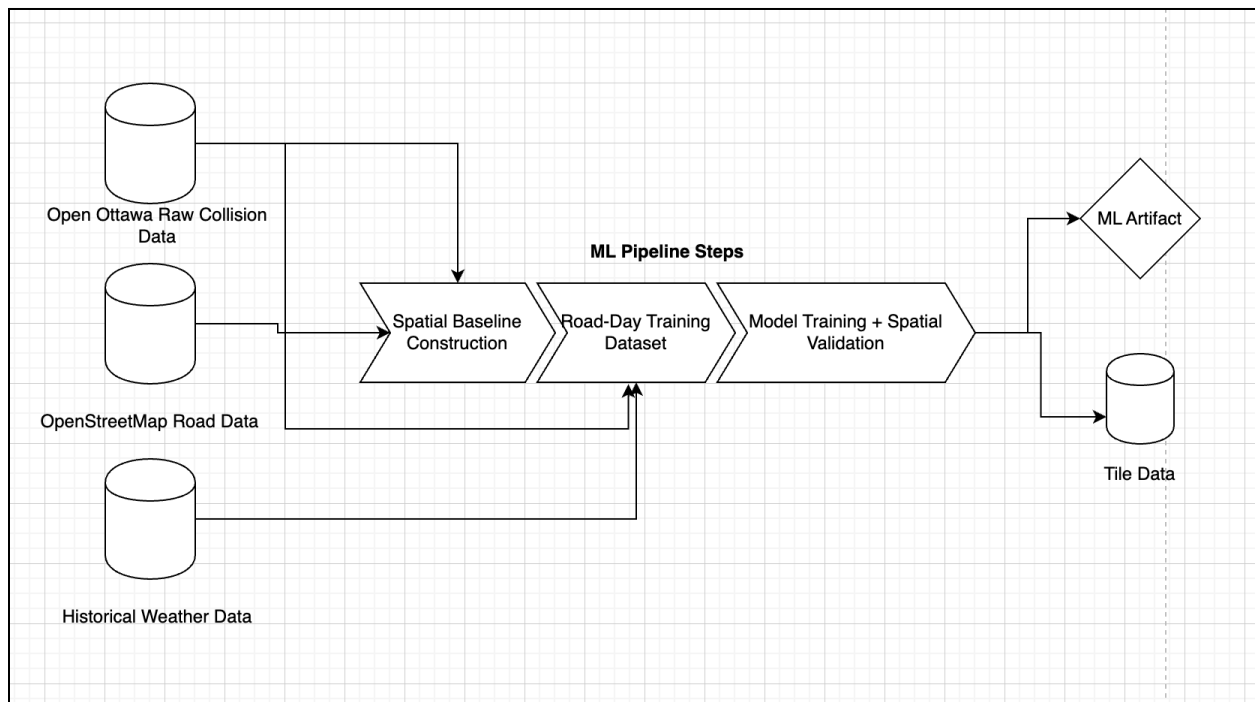
The interaction flow is simple:

1. The frontend loads the map and current Ottawa weather.
2. The user keeps live weather or selects a historical date.
3. Clicking a neighborhood loads nearby road segments.
4. The backend scores those segments with the trained XGBoost model.
5. Clicking a scored road opens a side panel with road statistics, current weather, model score, and an AI explanation.

## High-Level System Architecture



## High-Level ML Data Pipeline



## Main Components

- Data/: raw and processed project data
- data-pipeline/: offline spatial processing, feature engineering, tile generation, and training
- backend/: FastAPI inference service
- frontend/: React and MapLibre user interface

### 3. Project Evolution

#### 3.1 Initial MVP

The initial MVP learned mostly from historical collision count. That created an important problem: roads with very different real-world danger could both be flagged as high risk simply because they had recorded collisions.

For example, a downtown road such as Wellington Street with 8 collisions and a highway segment with 120 collisions could both appear high risk. This was misleading because the model was learning traffic volume more than danger.

The original baseline produced:

- ROC-AUC: 0.741

This looked strong, but the task was easier and less defensible. The model was closer to identifying historically busy crash corridors than forecasting contextual risk.

#### 3.2 Fixing the Denominator Problem

The first major improvement was solving the denominator problem. Since direct traffic exposure data such as AADT was not available for every road, the pipeline was redesigned to estimate relative risk using road length and road class as proxies.

Instead of relying on raw collision count, the system computed collisions per kilometer and compared each segment against other roads of the same OpenStreetMap highway class. This meant local streets were compared against local streets, and highways against highways.

#### 3.3 Moving to Spatiotemporal Forecasting

The next major improvement was adding weather. This introduced another challenge: the raw collision dataset only contains days when crashes happened. If weather were merged directly into those rows, the model would only learn what dangerous conditions look like, but not what safe days look like.

To fix this, the project was rebuilt as a road-day prediction problem:

- positive samples are real road-days with collisions
- negative samples are matched non-collision days on the same road
- daily Ottawa weather is merged into each road-day

This second iteration produced:

- ROC-AUC: 0.626

Although lower than the MVP, this was a more realistic result because the task and evaluation were both harder.

#### 3.4 Current Final System

The current version improves the system further by combining:

- normalized segment risk
- weather variables
- richer historical segment-level features
- connected-road context
- block spatial cross-validation
- live and historical weather scenarios
- local area scoring for frontend performance
- local LLM explanations for interpretability

The current model output is:

- Mean ROC-AUC: 0.6164
- Mean Average Precision: 0.3510

This lower headline ROC-AUC should not be interpreted as a weaker project. It reflects a harder and more honest problem setup. The final system is stronger because it is solving a better question under stricter evaluation.

## 4. Methodology

### 4.1 Spatial Baseline

Collision points are snapped to Ottawa road segments from OpenStreetMap. The baseline does not treat raw crash count as direct risk. Instead, it computes relative risk density so that each road is judged against others of the same class.

This matters because a highway and a residential road should not be compared by absolute crash count alone.

### 4.2 Road-Day Learning Setup

The learning problem is defined at the road-day level, not the crash-row level.

- positive rows: road-days with one or more collisions
- negative rows: matched non-collision days on the same segment

This better matches the real prediction task and prevents weather from being learned only from crash days.

### 4.3 Weather Integration

Historical daily weather for Ottawa is pulled from Meteostat and merged into the road-day matrix. The frontend uses live or historical weather from Open-Meteo during inference, allowing scenario-based testing.

### 4.4 Engineered Features

The current model uses more than road type and baseline risk. It includes:

- segment length and normalized baseline risk

- historical crash frequency
- connected-road context
- traffic control and intersection patterns
- wet, snow, ice, and freezing-rain tendencies
- dark-condition tendency
- vulnerable road-user involvement
- crash-type and severity patterns

## 4.5 Model Training

The model is trained with XGBoost and evaluated using 2 km block spatial cross-validation with 5 folds. This avoids random geographic leakage and gives a more realistic estimate of generalization to unseen areas.

## 5. Engineering

### 5.1 Backend

The backend loads the trained model, a per-segment feature lookup table, and exported threshold metadata. It exposes three endpoints:

| Endpoint        | Purpose                                              |
|-----------------|------------------------------------------------------|
| /model-metadata | returns thresholds and evaluation summary            |
| /predict-batch  | scores the road segments in a clicked local area     |
| /insight        | generates a short explanation for a selected segment |

### 5.2 Frontend

The frontend uses React and MapLibre. It loads a neutral road baseline first, then performs local scoring only when the user clicks an area. This avoids scoring the full city at once and keeps the app responsive.

It also supports:

- toggle-off behavior when the same active area is clicked
- road inspection after scoring
- live weather mode
- historical weather replay
- stats-first display followed by AI explanation

### 5.3 Reliability and Fallbacks

The system handles several failure cases:

- if weather loading fails, the app should not crash

- if backend requests fail, the map should still remain usable
- if Ollama is unavailable, the road stats panel still works even if the explanation is missing
- the tile server and Ollama setup must be documented clearly

## 6. Evaluation and Results

### 6.1 Why the Scores Changed

One of the most important points in this project is that the model score became lower as the methodology became better.

That happened because:

- the early MVP relied heavily on collision count
- later versions solved a harder forecasting problem
- spatial block cross-validation is stricter than easier random evaluation

So the final score should be interpreted as more credible, not worse.

### 6.2 Evaluation Setup

Evaluation uses:

- 2 km spatial blocks
- 5-fold GroupKFold
- geographically unseen regions at test time

This makes the results academically stronger because the model cannot simply memorize nearby road structure.

### 6.3 Main Comparison

| Iteration            | Main idea                                                                    | Main limitation                           | Result                              |
|----------------------|------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------|
| Initial MVP          | historical-count-based road risk                                             | confused volume with danger               | ROC-AUC 0.741                       |
| Second iteration     | normalized risk + weather-aware road-day modeling                            | still less feature-rich than final system | ROC-AUC 0.626                       |
| Current final system | richer road features + connected context + block spatial CV + live inference | harder task, stricter evaluation          | Mean ROC-AUC 0.6164, Mean AP 0.3510 |

## 6.4 Why Average Precision Matters

This is an imbalanced classification problem, so Average Precision is important. ROC-AUC shows ranking quality overall, but Average Precision better reflects how well the model retrieves genuinely risky road-days when positives are much rarer than negatives.

## 6.5 Current Cross-Validation Results

| Fold | ROC-AUC | Average Precision |
|------|---------|-------------------|
| 1    | 0.6169  | 0.3517            |
| 2    | 0.6131  | 0.3474            |
| 3    | 0.6156  | 0.3481            |
| 4    | 0.6206  | 0.3564            |
| 5    | 0.6159  | 0.3516            |
| Mean | 0.6164  | 0.3510            |

## 6.6 Feature Interpretation

The top learned features in the current model include:

- rain\_environment\_rate
- snow\_environment\_rate
- historical\_crash\_day\_count\_log1p
- tavg
- prcp
- historical\_event\_count\_log1p

This supports the idea that the final model is not relying on raw collision count alone. It is using a mix of weather and historical segment behavior.

## 7. Limitations

Important limitations remain:

- the raw dataset appears to be missing 2023
- the project does not include direct traffic volume or AADT
- OpenStreetMap segmentation may fragment roads unnaturally
- daily weather is only a partial proxy for true real-time conditions
- there is no prospective future-year validation yet
- AI explanations depend on local LLM behavior

## 8. Literature Review and Research Context

This project is informed by prior work in geospatial machine learning, road safety modeling, weather-aware prediction, and explainable AI. The literature mainly supports four design choices in this report: using spatially valid evaluation, avoiding raw collision totals as a direct measure of danger, including weather as part of the prediction problem, and presenting results in an



interpretable way. The following short review summarizes the most relevant ideas and shows how they shaped the final system.

### **8.1 Spatial Leakage**

This project is based on the idea that spatial data should not be evaluated like ordinary tabular data. Tobler's First Law says that nearby things are more related than distant things, which is why spatial autocorrelation appears in geographic datasets (Tobler, 1970). ArcGIS documentation on Moran's I explains that when nearby features have similar values, the pattern is spatially clustered rather than random. Because of this, random train/test splits can be misleading in geospatial machine learning. Roberts et al. show that cross-validation should account for spatial structure, and Spatial+ also shows that random validation can give over-optimistic results in geospatial models (Roberts et al., 2017; Wang et al., 2023). This supports the use of 2 km block spatial cross-validation in this project.

### **8.2 Crash Risk Modeling**

Road safety research shows that raw crash totals are not the same as true risk. A road with more crashes may simply carry more traffic, while a road with fewer crashes may still be more dangerous relative to its size or type. Zhang et al. show that crash frequency behaves differently across urban roadway segment types, which supports comparing roads in a more normalized way instead of using raw totals alone (Zhang et al., 2012). This is why this project moved away from treating collision count as the main signal and instead used normalized segment risk.

### **8.3 Weather-Aware Prediction**

Previous studies show that weather conditions such as rain, snow, and freezing precipitation can increase crash risk. Eisenberg found that precipitation changes crash patterns, and Tobin et al. showed that different precipitation types can have different relative effects on crash risk (Eisenberg, 2004; Tobin et al., 2021). This supports the decision to move beyond a static collision map and model road-segment risk under changing weather conditions.

### **8.4 Negative Sampling**

One challenge in this project is that the raw collision dataset only records days when crashes happened. That creates a biased learning problem unless non-collision examples are also included. Matched comparison designs are commonly used when positive events are rare and a meaningful contrast group is needed (Bansal & Pepe, 2013). This supports the use of matched safe-day negatives in this project so the model can learn both risky and non-risky road-day conditions.

### **8.5 Explainability**

This project is also influenced by work on explainable AI, which argues that systems should help users understand why a prediction was made rather than only showing a score (Minh et al., 2022). That is especially relevant in a safety setting, where users benefit from interpretable results instead of just raw model output. This supports the decision to include an AI explanation panel in the final system.

## 8.6 Project Contribution

This project does not propose a new machine learning algorithm. Its contribution is the integration of research-supported ideas into one working system: normalized spatial risk, weather-aware road-day prediction, spatially valid evaluation, interactive local inference, efficient vector-tile map delivery, and explanation generation in a usable interface. Vector tiles are especially useful here because they are lightweight, fast to render, and support interactive querying in web maps. Together, these choices make the project a strong end-to-end application grounded in relevant research.

## 9. Conclusion

This project started as a static collision map and developed into a context-aware traffic risk forecasting system. The final version is stronger than the initial MVP not because it produces a higher easy score, but because it solves a harder and more meaningful problem. By combining normalized road risk, weather-aware modeling, block spatial validation, live inference, and explainable interaction, the project demonstrates both technical ambition and academic rigor.

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